

Analysis of cross-orientation responses as a function of depth in primary visual cortex in macaque, a two-photon imaging study

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Abstract

1 Introduction

Cross-orientation suppression is a hallmark of response normalization in primary visual cortex (V1): when two gratings at orthogonal orientations are superimposed to form a plaid, the response of an orientation-selective neuron typically falls below the linear sum of its responses to each grating alone (Morrone et al., 1982; Bonds, 1989; Carandini et al., 1997; Freeman et al., 2002). This phenomenon, studied for over fifty years since the early descriptions of orientation selectivity (Hubel and Wiesel, 1962, 1968), has been central to understanding how cortical circuits combine and normalize visual inputs.

Several mechanistic accounts of cross-orientation suppression have been proposed. Intracortical feedback inhibition models posit that orientation-tuned inhibitory interneurons directly suppress

responses to non-preferred orientations (Sillito, 1975; Bonds, 1989; Shapley et al., 2003). Feedforward inhibition models attribute suppression to broadly tuned inhibition arriving from the lateral geniculate nucleus (Priebe and Ferster, 2006; Li et al., 2006). Contrast saturation and divisive normalization models explain suppression as a consequence of a compressive nonlinearity in the contrast response function, in which the summed driving input from both gratings falls on the saturating portion of the neuron’s response curve (Heeger, 1992; Albrecht and Hamilton, 1982; Carandini and Heeger, 2012). These mechanisms are not mutually exclusive, and their relative contributions likely depend on cortical circuit architecture.

Divisive normalization, in particular, has emerged as a canonical computation underlying cross-orientation suppression (Heeger, 1992; Carandini and Heeger, 2012). In the normalization model, each neuron’s response is divided by a weighted sum of activity across a pool of neurons, producing suppression whenever the pooled activity is large. This framework accounts quantitatively for cross-orientation suppression in both simple and complex cells (Carandini et al., 1997), explains contrast gain control (Albrecht and Hamilton, 1982; Sclar et al., 1990; Schwartz and Simoncelli, 2001), and generalizes to attentional modulation (Reynolds and Heeger, 2009).

Nearly all prior work on cross-orientation suppression has relied on single-unit electrophysiology, recording from one neuron at a time. Although this approach has yielded precise characterizations of suppression in individual cells (DeAngelis et al., 1992; Carandini et al., 1997; Smith et al., 2006; Busse et al., 2009), it leaves open how suppression and facilitation are distributed across neural populations and whether they vary systematically with cortical depth. Moreover, the majority of experiments have been conducted in cat V1, with comparatively fewer studies in macaque (Carandini et al., 1997; Smith et al., 2006), the species whose cortical organization most closely resembles that of humans.

Layer 2/3 of macaque V1 spans several hundred micrometers and contains sublayers with distinct connectivity patterns, cell-type distributions, and functional properties (Lund, 1988; Callaway, 1998; Douglas and Martin, 2004; Briggs, 2010). Superficial portions of layer 2/3 receive strong input from layer 4C and participate in local recurrent circuits, while deeper portions connect more extensively with layer 5 and other cortical layers (Callaway, 1998; Douglas and Martin, 2004). Whether cross-orientation interactions change systematically across this depth range has not been directly tested at the population level.

The functional architecture of macaque V1 is organized into orientation columns and pinwheel structures, within which nearby neurons share similar orientation preferences (Blasdel and Salama, 1986; Bonhoeffer and Grinvald, 1991; Ohki et al., 2005, 2006). Local homogeneity of orientation preference, spatial frequency tuning, and orientation selectivity all contribute to the local circuit context in which cross-orientation interactions occur (Nauhaus et al., 2008; Ringach et al., 2002). How these functional properties covary with depth and interact with normalization remains poorly understood.

Two-photon calcium imaging provides the combination of cellular resolution and population-scale recording needed to address these questions (Denk et al., 1990; Stosiek et al., 2003; Grienberger and Konnerth, 2012). Recent advances in genetically encoded calcium indicators (Chen et al., 2013) and viral delivery strategies for nonhuman primates (Deverman et al., 2016; Li et al., 2017; Tang et al., 2018) have made it possible to record simultaneously from hundreds of neurons at identified cortical depths.

Here, we used two-photon calcium imaging with GCaMP6s to measure cross-orientation interactions across approximately 400 μm of cortical depth in layer 2/3 of macaque V1. We recorded from nearly 5,000 neurons across 28 fields of view spanning 140–518 μm below the pial surface. For each neuron, we compared the observed response to orthogonal plaid stimuli with a linear prediction derived from single-grating tuning curves, quantifying suppression and facilitation with two complementary metrics. We find that cross-orientation interactions transition systematically from facilitation in superficial layer 2/3 to suppression at greater depth, consistent with stronger normalization in deeper sublayers. This depth gradient is partially mediated by spatial frequency preference and orientation tuning bandwidth, both of which also vary with depth.

The long-term goal of this project is to combine large-scale functional imaging with dense connectomics to relate circuit wiring to population computation in primate V1 (Chatterjee et al., 2021). By targeting wide cortical blocks, the physiology of as many neurons as possible can be compared post-mortem to serial electron microscopy reconstructions from the same tissue.

2 Methods

2.1 Neurophysiology

Two-photon calcium imaging was performed in primary visual cortex (V1) of one anesthetized pigtail macaque (*Macaca nemestrina*). The calcium indicator GCaMP6s was delivered via injection of the viral vector PHP.eB-CAG-GCaMP6s into V1. A cranial imaging window was implanted over the injection site, and the animal was allowed a 26-day recovery period for viral expression. Terminal recording sessions were conducted over 5 consecutive days under general anesthesia with standard physiological monitoring.

Two-photon imaging was performed using a modified Sutter MOM (Movable Objective Microscope) system equipped with a Spectra-Physics MaiTai Ti:sapphire laser. Image acquisition was controlled by ScanImage software. Single-plane scanning was performed at 30 Hz frame rate. Laser power was adjusted for each imaging depth to compensate for increased scattering at greater depths while staying below the threshold for photodamage.

A total of 28 fields of view (FOV) were acquired at depths ranging from 140 to 518 μm below the pial surface, spanning the extent of layer 2/3. Each FOV covered a 512×512 pixel area. Regions of interest (ROIs) corresponding to individual neurons were identified using Suite2p (Pachitariu et al., 2017), yielding approximately 4,785 ROIs across all sites (mean $\pm$ SD: 171 ± 28 ROIs per site; range: 120–230).

2.2 Visual stimuli

Visual stimuli consisted of drifting sinusoidal gratings and orthogonal plaids. Gratings were presented at 50% contrast, 4 Hz temporal frequency, 4 cyc/deg spatial frequency, within a 2-degree circular patch centered on the aggregate receptive field of each recording site. Plaid stimuli were constructed by superimposing two gratings at orthogonal orientations, each at the same contrast, temporal frequency, and spatial frequency as the component gratings.

Stimulus direction was varied in 30-degree increments (12 directions spanning 0–330°). Each stimulus was presented for 2 s, followed by a 1.5-s mean-gray inter-stimulus interval. Each direction condition was repeated for 6 trials, and trial order was randomized within each block.

In addition to the grating and plaid stimuli, receptive field maps were obtained using flashed

light and dark spots, and spatial frequency tuning was measured at 8 spatial frequencies (0.64–7.68 cyc/deg).

2.3 Data analysis

Image registration and ROI extraction were performed using Suite2p (Pachitariu et al., 2017). For each ROI, the stimulus-evoked response was computed as the mean fluorescence in the window 200–2,000 ms after stimulus onset. The baseline fluorescence was computed from the window 250–550 ms before stimulus onset (during the preceding inter-stimulus interval). All analyses were performed on baseline-subtracted responses (ΔF).

2.3.1 Metric P: suppression/facilitation ratio

For each ROI, a linear prediction of the plaid response was constructed as follows. The baseline-subtracted grating tuning curve $G(\theta)$ was shifted by -90° and summed with the original to produce the predicted plaid tuning curve:

$$\text{Pred}(\theta) = [G(\theta) - B] + [G(\theta - 90^\circ) - B], \quad (1)$$

where B is the baseline fluorescence. The primary metric P quantifies the ratio of the observed plaid response to this linear prediction, summed across all directions:

$$P = \frac{\sum_i R_i^{\text{plaid}}}{\sum_i \text{Pred}_i}, \quad (2)$$

where R_i^{plaid} is the baseline-subtracted plaid response at direction i . Values of $P > 1$ indicate facilitation (the observed plaid response exceeds the linear prediction), $P < 1$ indicates suppression (the observed response falls below the prediction), and $P = 1$ indicates perfect linearity. Because P is a ratio, it is plotted on a logarithmic axis so that equal distances represent equal multiplicative changes and the neutral point ($P = 1$) falls at the center of the scale.

2.3.2 Metric C: shape similarity

Metric C is the Pearson correlation between the predicted and observed plaid tuning curves:

$$C = r(\text{Pred}(\theta), R^{\text{plaid}}(\theta)). \quad (3)$$

Higher values of C indicate that the shape of the plaid tuning curve is more linearly predictable from the component grating responses, independent of the overall amplitude ratio captured by P .

2.3.3 Signal-to-noise ratio

Signal-to-noise ratio (SNR) was computed separately for grating and plaid responses as:

$$\text{SNR} = \frac{\text{Var}_{\text{signal}}}{\text{Var}_{\text{noise}}}, \quad (4)$$

where signal variance was estimated from the tuning curve means across directions and noise variance from the trial-to-trial variability.

2.3.4 Additional metrics

Orientation Selectivity Index (OSI). OSI was computed using standard circular variance methods (Ringach et al., 2002) to quantify the degree of orientation tuning for each neuron.

Best spatial frequency. The spatial frequency eliciting the maximum response was determined from tuning curves measured at 8 spatial frequencies (0.64–7.68 cyc/deg) (De Valois et al., 1982).

Orientation tuning half-width. Half-width at half-height of the grating tuning curve, measured on the flank toward the orthogonal orientation. This captures how broadly the neuron responds toward the orientation of the cross-oriented grating component.

Local Homogeneity Index (LHI). LHI quantifies the similarity of orientation preference between neighboring ROIs within each field of view, computed using a Gaussian weighting kernel with $\sigma = 35 \mu\text{m}$ (Nauhaus et al., 2008). Higher LHI indicates more homogeneous local orientation preference, as found near orientation domain centers; lower LHI indicates heterogeneous preferences,

as found near pinwheel centers.

2.4 Statistical analysis

2.4.1 Site-level analysis

For primary analyses, metrics were averaged across all ROIs within each field of view to obtain site-level means ($n = 28$). This approach accounts for the hierarchical structure of the data (ROIs nested within sites) and avoids pseudoreplication.

2.4.2 Correlation and regression

Pearson correlation coefficients were computed between cortical depth and metric values. Statistical significance was assessed using two-tailed tests with $\alpha = 0.05$. Linear regression was used to estimate depth slopes. Non-parametric bootstrap (10,000 iterations, resampling sites with replacement) was used to compute 95% confidence intervals for correlation coefficients.

2.4.3 Partial correlations

To control for potential confounds, partial correlations were computed between depth and metrics while statistically controlling for ROI radius, SNR, OSI, and best spatial frequency. Partial correlations were obtained by regressing out confound variables from both depth and metric values, then correlating the residuals.

2.4.4 Mixed-effects models

Linear mixed-effects models were fit using restricted maximum likelihood (REML) with recording site as a random intercept:

$$\text{Metric}_{ij} = \beta_0 + \beta_1 \cdot \text{Depth}_j + \text{Covariates} + u_j + \varepsilon_{ij}, \quad (5)$$

where $u_j \sim \mathcal{N}(0, \sigma_u^2)$ is the random intercept for site j . This approach accounts for the non-independence of ROIs within sites while leveraging the full ROI-level dataset ($n = 4,785$).

2.4.5 Mediation analysis

To identify which covariates mediate the P -depth relationship, partial correlations were computed controlling for each covariate individually. The percent reduction in correlation strength relative to the zero-order correlation quantifies the mediating influence of each variable.

2.4.6 Subpopulation analyses

Depth-metric correlations were computed separately for subpopulations defined by median splits on OSI and spatial frequency preference, and by OSI quartiles. Fisher's z -transformation was used to compare correlation coefficients between low-OSI (Q1) and high-OSI (Q4) subpopulations.

3 Results

3.1 Dataset overview

We recorded calcium signals from 4,785 ROIs across 28 fields of view in layer 2/3 of macaque V1, spanning cortical depths of 140–518 μm below the pial surface (Figure 1). Each ROI was presented with drifting sinusoidal gratings and orthogonal plaids, and cross-orientation interaction metrics were computed by comparing the observed plaid response to a linear prediction derived from the grating tuning curve.

3.2 Cross-orientation suppression increases with cortical depth

Metric P showed a strong negative correlation with cortical depth at the site level ($r = -0.706$, $p = 2.72 \times 10^{-5}$, $n = 28$; Figure 3A). Superficial sites (140–266 μm) exhibited $P > 1$, indicating facilitation—the observed plaid response exceeded the linear prediction. Deeper sites (392–518 μm) exhibited $P < 1$, indicating suppression. The transition from facilitation to suppression occurred at approximately 350 μm depth.

3.3 Plaid tuning shape becomes more predictable with depth

Metric C , the Pearson correlation between predicted and observed plaid tuning curves, showed a strong positive correlation with depth ($r = +0.798$, 95% CI: $[+0.675, +0.891]$, $p = 3.67 \times 10^{-7}$;

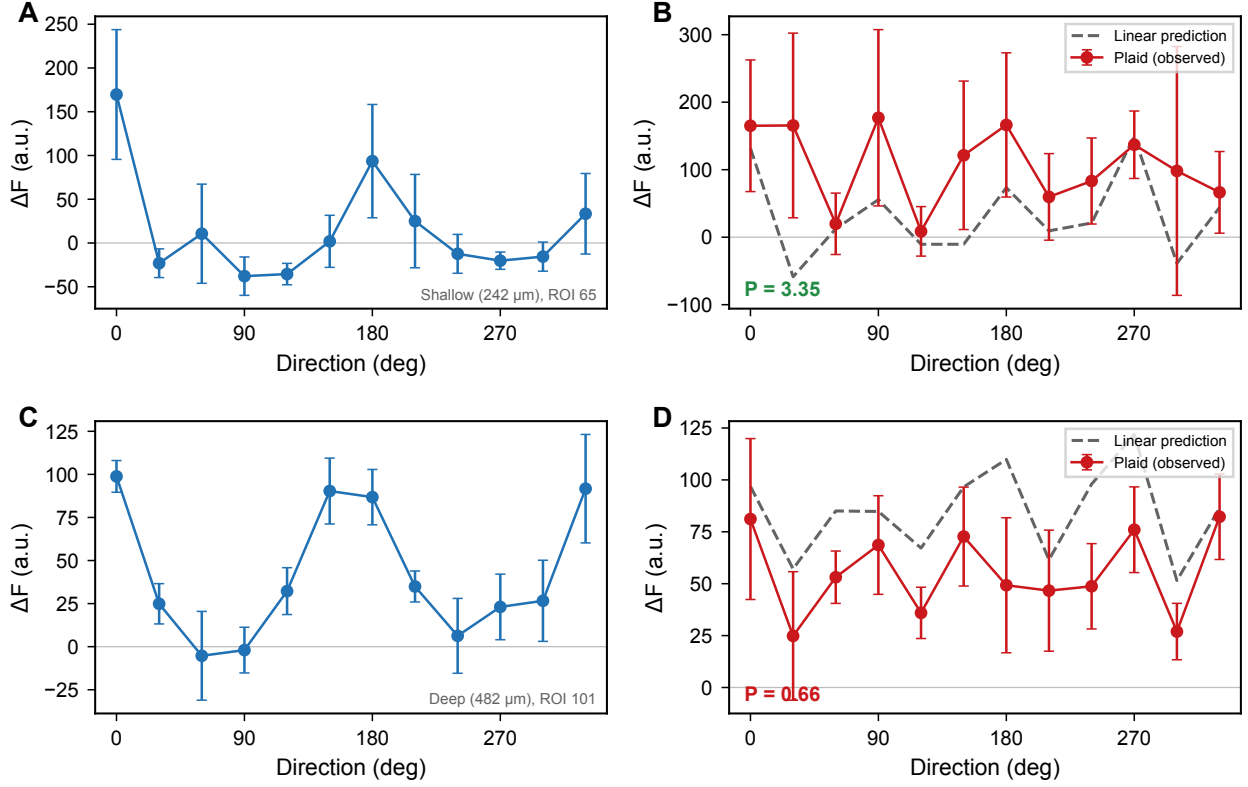


Figure 1: **Example cross-orientation analysis at shallow and deep cortical depths.** (A) Grating tuning curve (ΔF) for an example ROI at a shallow site (242 μm). (B) Observed plaid response (red) versus linear prediction (dashed gray) for the same ROI. $P = 3.35$, indicating strong facilitation. (C) Grating tuning curve for an example ROI at a deep site (482 μm). (D) Observed plaid response versus prediction for the deep ROI. $P = 0.66$, indicating suppression. Error bars show SEM across trials.

Figure 2: **Linear prediction model for plaid responses.** (A) Gabor spatial filter at the preferred orientation. (B) Temporal filter modeled as a difference of Maxwell functions (time constants $s_1 = 20$ ms, $s_2 = 40$ ms), producing a space-time separable linear filter. (C) The linear prediction is the sum of the responses to each component grating of the plaid, computed by shifting and summing the grating tuning curve. Deviations from this prediction are quantified by metrics P (ratio) and C (shape correlation).

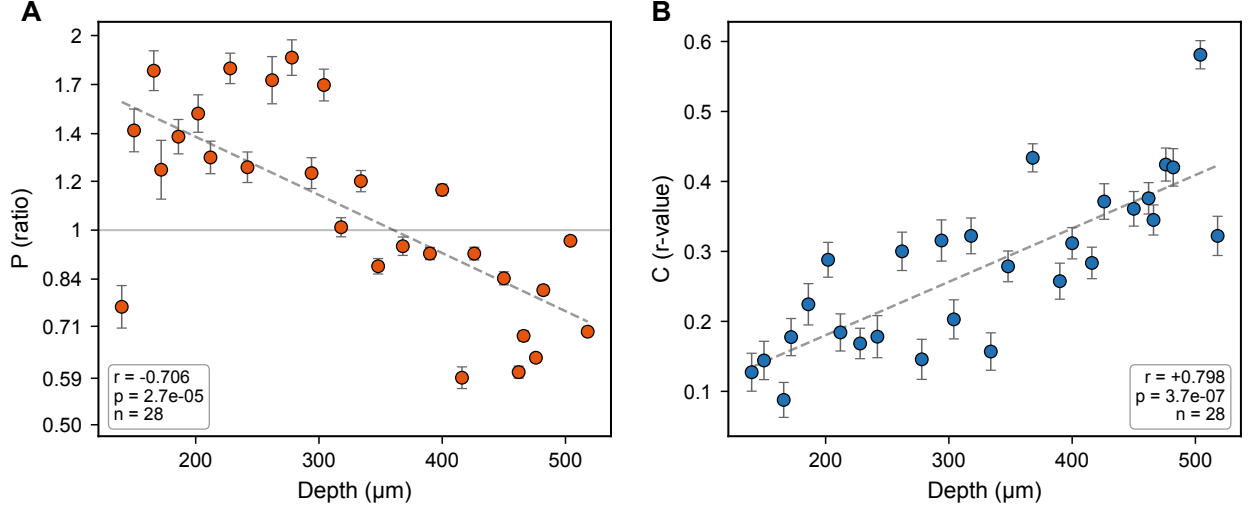


Figure 3: **Depth-dependent changes in cross-orientation metrics.** (A) Metric P (suppression/facilitation ratio) decreased with cortical depth ($r = -0.706$, $p = 2.72 \times 10^{-5}$, $n = 28$). Each point represents the site-mean P across all ROIs at one imaging depth, plotted on a logarithmic axis. Dashed line shows linear regression. $P > 1$: facilitation; $P < 1$: suppression. (B) Metric C (shape similarity) increased with depth ($r = +0.798$, $p = 3.67 \times 10^{-7}$). Higher C indicates the plaid tuning curve shape is more linearly predictable from grating responses. (C) Scatter plot of P versus C across sites, colored by cortical depth, showing their anticorrelation. (D) Depth-binned summary. Superficial layer 2/3 shows facilitation ($P > 1$); deep layer 2/3 shows suppression ($P < 1$).

Figure 3B). This indicates that deeper neurons, despite showing stronger suppression (lower P), have plaid tuning curve shapes that are more linearly predictable from their grating responses. The metrics P and C were anticorrelated across sites ($r = -0.647$, $p < 0.001$; Figure 3C).

3.4 Spatial frequency preference varies with depth

Mean preferred spatial frequency was higher at superficial depths and decreased with cortical depth (Figure 4). Within individual sites, spatial frequency showed weak correlations with P (significant at 7 of 28 sites), indicating that spatial frequency preference does not strongly predict P within a single imaging plane. However, across the full depth range, mediation analysis revealed that spatial frequency preference accounted for approximately 40% of the P -depth correlation: the partial correlation between P and depth controlling for spatial frequency was $r = -0.424$ (compared to $r = -0.706$ at zero order; Table 1).

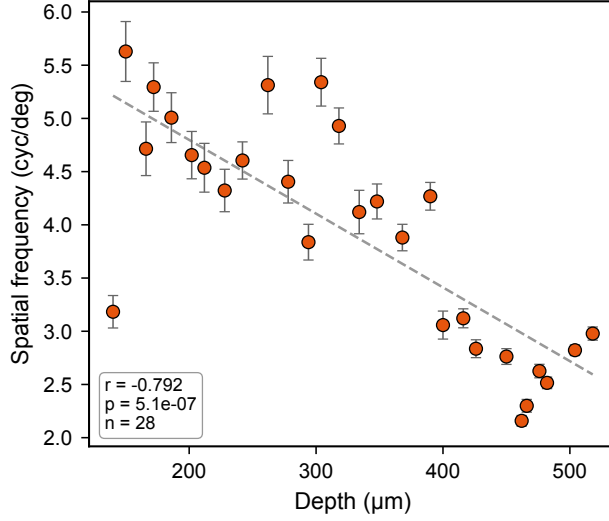


Figure 4: **Spatial frequency preference decreases with cortical depth.** Site-mean preferred spatial frequency plotted against depth. Each point is one field of view; error bars show SEM. Dashed line is the linear regression fit.

Figure 5: **Baseline fluorescence increases with depth.** Site-mean baseline fluorescence as a function of cortical depth, reflecting the increase in laser power at deeper imaging planes.

3.5 Baseline fluorescence increases with depth

Baseline fluorescence increased with cortical depth, consistent with the progressive increase in laser power applied at deeper imaging planes to compensate for optical scattering (Figure 5). Within individual sites, P and baseline fluorescence were uncorrelated at 24 of 28 sites, indicating that the depth gradient in P is not an artifact of imaging depth. All metrics used baseline-subtracted responses.

3.6 Orientation tuning bandwidth broadens with depth

Orientation tuning half-width increased with cortical depth ($r = +0.869$, $p = 2.04 \times 10^{-9}$; Figure 6A), with mean half-width broadening from approximately 60° at superficial sites to approximately 80° at deeper sites. Within individual sites, half-width showed a significant negative correlation with C at 15 of 28 sites (Figure 6B): at a given depth, more broadly tuned neurons tended to have less linearly predictable plaid tuning shapes. Tuning bandwidth mediated approximately 21% of the P -depth effect (Table 1).

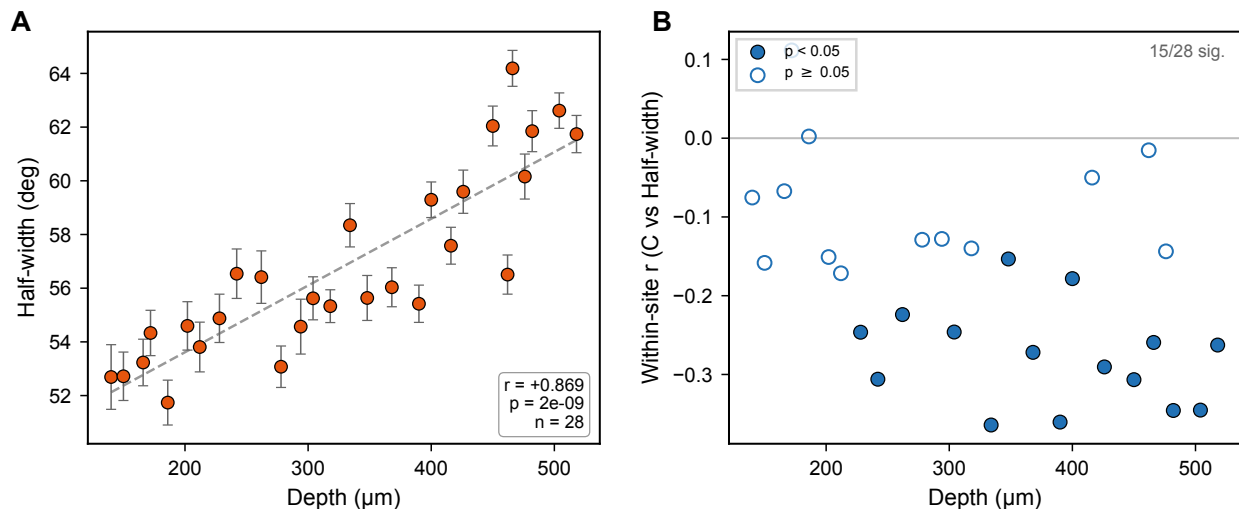


Figure 6: **Orientation tuning bandwidth increases with depth.** (A) Site-mean half-width at half-height versus depth ($r = +0.869$, $p = 2.0 \times 10^{-9}$). (B) Within-site Pearson r between half-width and C , plotted against depth. Filled circles: $p < 0.05$; open circles: $p \geq 0.05$. The consistent negative correlation shows that, within sites, more broadly tuned neurons have less predictable plaid tuning shapes.

3.7 Local homogeneity decreases with depth

The local homogeneity index (LHI) showed a strong negative correlation with depth ($r = -0.93$, $p \ll 0.001$; Figure 7A), with LHI decreasing by approximately 35% from 150 to 520 μm . This is consistent with a transition from more homogeneous orientation preference maps in superficial layer 2/3 (orientation domains) to more heterogeneous maps at depth (near pinwheel centers or in regions with less columnar organization).

Within individual sites, C and LHI showed a consistently moderate negative correlation at 26 of 28 sites (Figure 7B), indicating that neurons in more heterogeneous orientation neighborhoods (lower LHI, near pinwheel centers) had more linearly predictable plaid tuning shapes.

3.8 Orientation selectivity mildly decreases with depth

OSI showed a mild decrease with cortical depth, from approximately 0.66 at 150 μm to 0.59 at 520 μm . Within individual sites, C and OSI were strongly positively correlated at 26 of 28 sites: neurons with higher orientation selectivity showed more linearly predictable plaid tuning. In contrast, P showed no significant relationship with OSI at 21 of 28 sites (Figure 8). Six deeper sites showed weak negative P –OSI correlations ($r \approx -0.2$ to -0.49), but the lack of relationship at

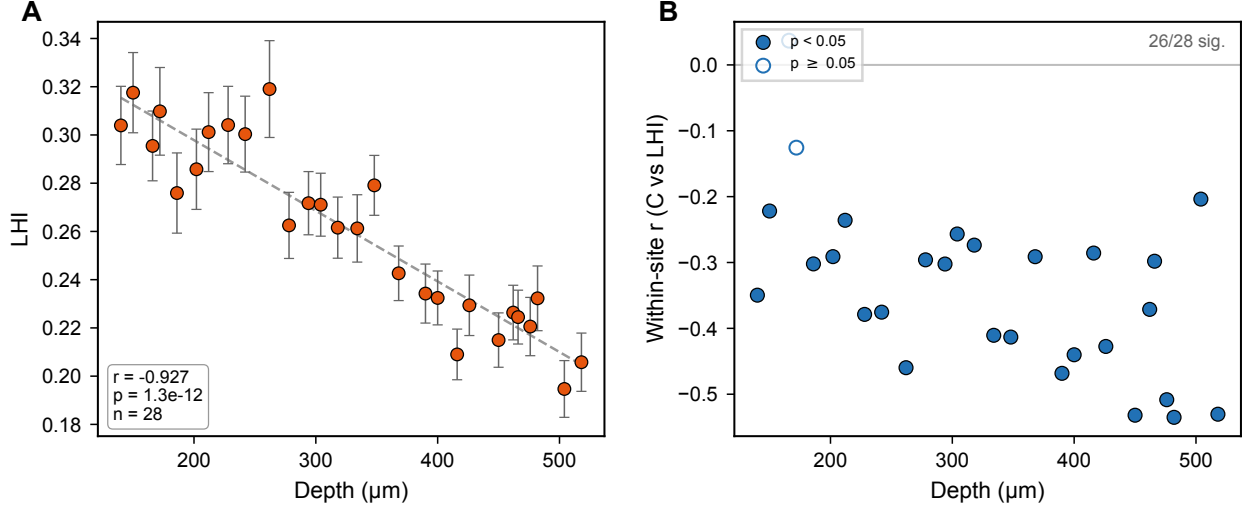


Figure 7: **Local homogeneity index decreases with depth.** (A) Site-mean LHI versus depth ($r = -0.93$). Higher LHI indicates more homogeneous local orientation preferences (orientation domains); lower LHI indicates greater heterogeneity (near pinwheels). (B) Within-site Pearson r between C and LHI, plotted against depth. A consistent negative relationship is observed at 26/28 sites.

shallow depths suggests that the strength of cross-orientation suppression is largely independent of orientation selectivity across most of layer 2/3.

3.9 Robustness and control analyses

The P -depth and C -depth correlations remained significant after controlling for ROI radius, SNR, OSI, and spatial frequency preference using partial correlations (Table 2). Mixed-effects models at the ROI level confirmed the P -depth relationship ($\beta_{\text{depth}} = -9.30$, $p = 7.9 \times 10^{-10}$; Table 3), and the effect survived additional controls for ROI radius and SNR ($\beta_{\text{depth}} = -8.64$, $p = 2.2 \times 10^{-8}$).

Subpopulation analyses confirmed that the P -depth effect was robust across neurons with high versus low OSI and high versus low spatial frequency preference.

The C -depth relationship was highly significant in the simple mixed-effects model ($\beta_{\text{depth}} = +0.086$, $p = 1.7 \times 10^{-11}$), but lost significance after controlling for ROI radius and SNR ($p = 0.151$), indicating partial confounding by morphological factors.

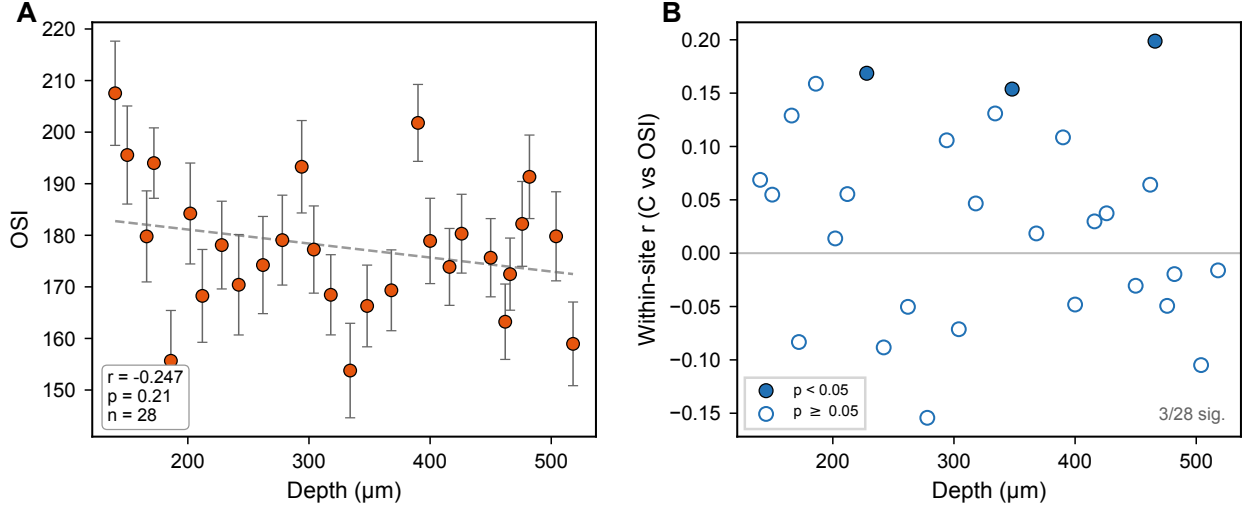


Figure 8: **Orientation selectivity and cross-orientation metrics.** (A) Site-mean OSI versus depth, showing a mild decrease. (B) Within-site Pearson r between C and OSI versus depth. A strong positive relationship is present at 26/28 sites: more orientation-selective neurons have more predictable plaid tuning shapes. (C) Within-site Pearson r between P and OSI versus depth. No consistent relationship at 21/28 sites.

Table 1: Mediation of the P -depth relationship by covariates.

Control variable	Partial r	% Reduction
None (zero-order)	-0.706	—
+ Spatial frequency	-0.424	40%
+ Half-width	-0.558	21%
+ LHI	-0.642	9%
+ SNR	-0.671	5%
+ OSI	-0.678	4%

Table 2: Partial correlations between depth and cross-orientation metrics after controlling for confounds.

Metric	Controls	Partial r	p -value
P	ROI radius	-0.751	< 0.001
P	All (radius + SNR + OSI + SF)	-0.520	0.005
C	ROI radius	+0.489	0.008
C	All (radius + SNR + OSI + SF)	+0.379	0.047

Table 3: Mixed-effects model results (ROI-level, $n = 4,785$, random intercept for site).

Model	β_{depth}	p -value
$P \sim \text{Depth} + (1 \text{Site})$	-9.30	7.9×10^{-10}
$P \sim \text{Depth} + \text{Radius} + \text{SNR} + (1 \text{Site})$	-8.64	2.2×10^{-8}
$C \sim \text{Depth} + (1 \text{Site})$	+0.086	1.7×10^{-11}
$C \sim \text{Depth} + \text{Radius} + \text{SNR} + (1 \text{Site})$	+0.036	0.151 (NS)

Table 4: Depth profile summary across layer 2/3.

Metric	Change with increasing depth
ROI size (radius, npix)	Increases
Baseline fluorescence	Increases
SNR	Increases
Orientation tuning bandwidth	Broader
Preferred spatial frequency	Lower
Orientation selectivity (OSI)	Lower
Local homogeneity (LHI)	Lower
Cross-orientation metric (P)	More suppressive ($P < 1$)

3.10 Summary of depth-dependent changes

Eight metrics showed coordinated depth-dependent changes across layer 2/3 (Table 4). Superficial layer 2/3 was characterized by high spatial frequency preference, narrow orientation tuning, high selectivity, high local homogeneity, and cross-orientation facilitation. Deeper layer 2/3 showed low spatial frequency preference, broad tuning, lower selectivity, lower local homogeneity, and cross-orientation suppression.

4 Discussion

Acknowledgments

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